THE LAWS OF SOFTWARE TEAMS

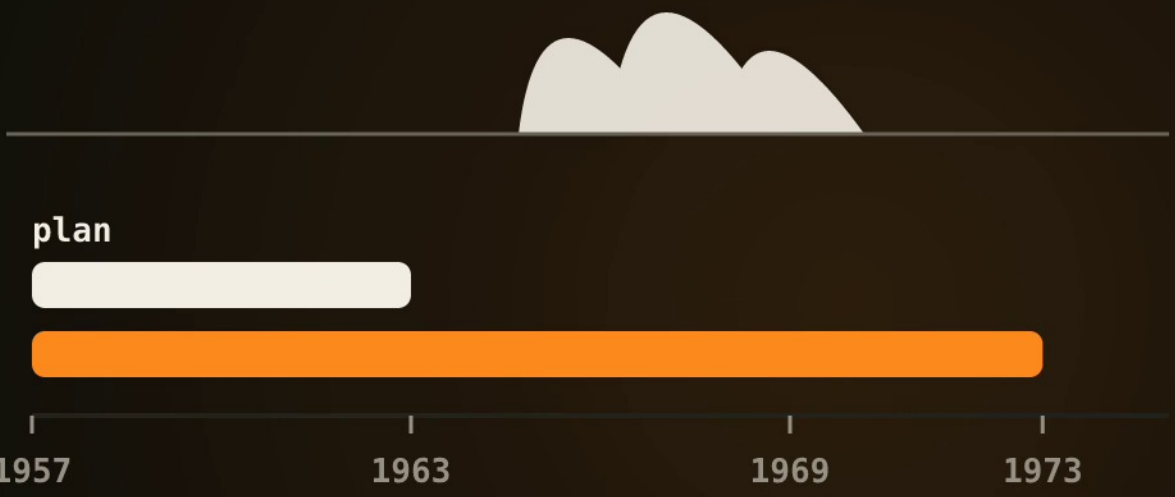
Hofstadter's law

It always takes longer than you expect, even when you take this law into account

swipe →

Sydney Opera House

Planned: 1963
Opened: 1973



cost **$102M**

10 years late · ~15× the budget

02 THE LAW

“It always takes longer than you expect, even when you take into account Hofstadter’s law”

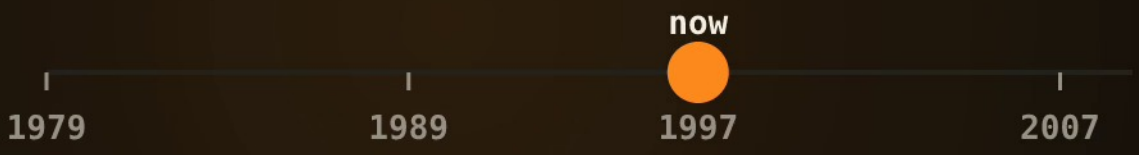


re-estimates: 3

Always ten years away

Douglas Hofstadter · Gödel, Escher, Bach · 1979

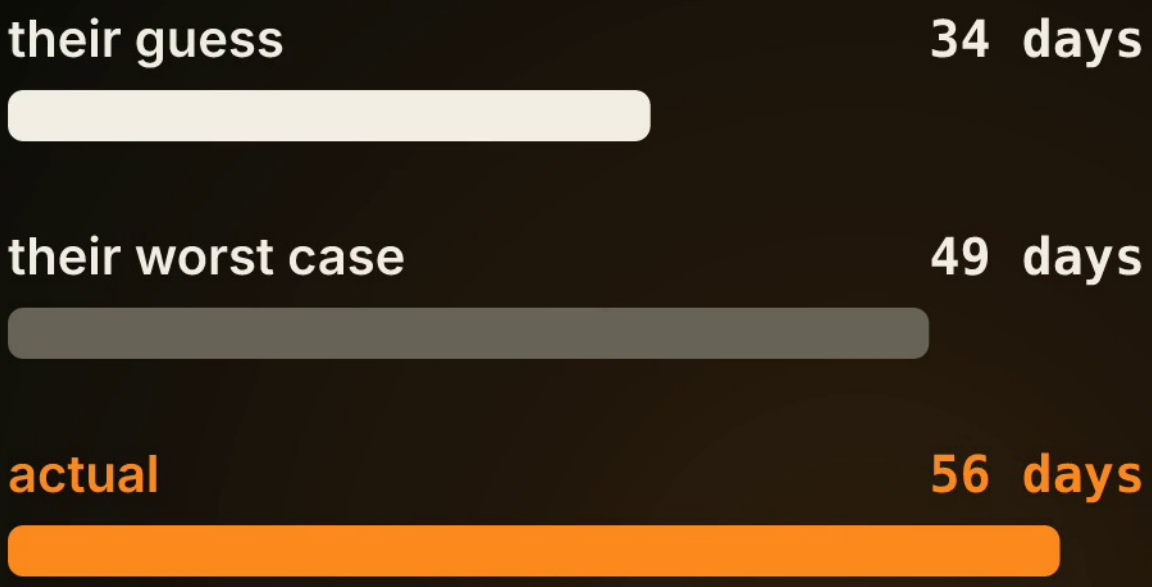
When will a computer beat the chess champion?



1997: Deep Blue beats Kasparov

Even the worst case was too optimistic

Students predicting when they'd finish a thesis



Only 30% finished by their own guess

Buehler, Griffin & Ross, 1994 · the "planning fallacy"

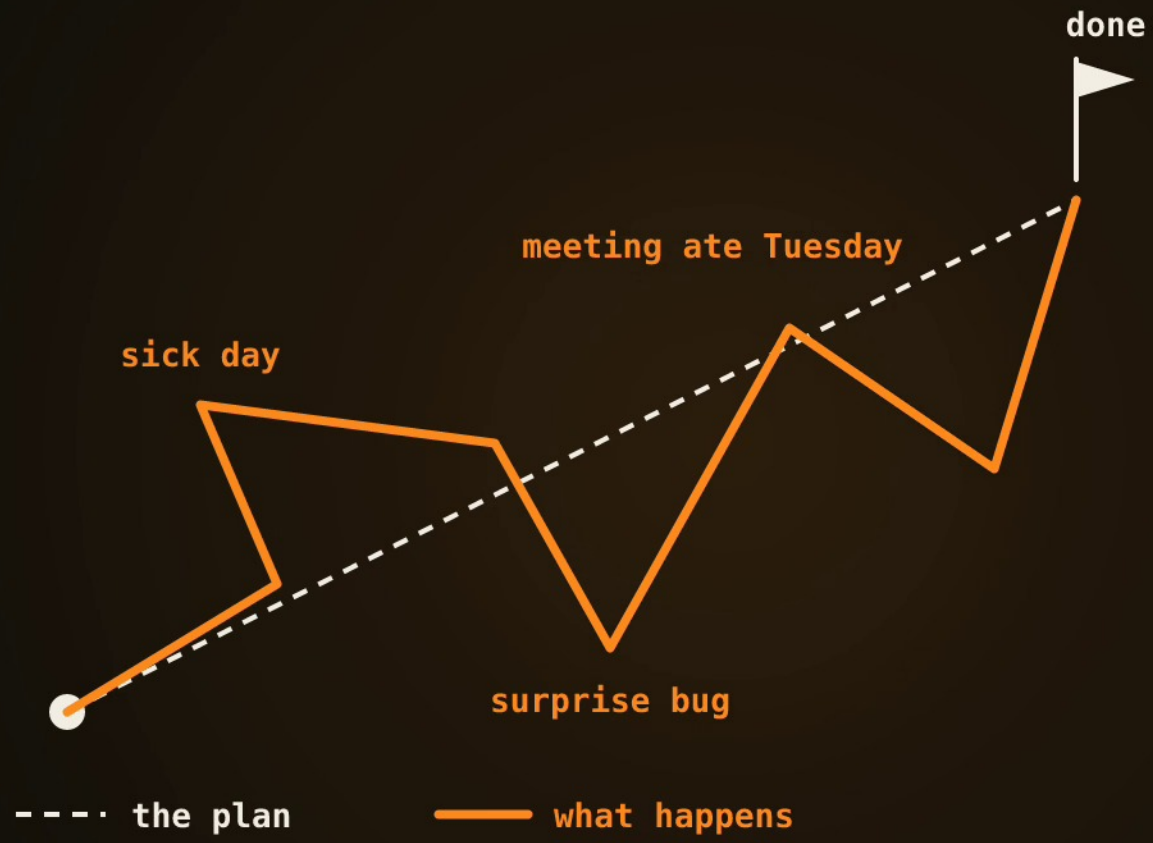
1 in 6 IT projects blows up



+200% cost **+70%** time

1,471 IT projects · Flyvbjerg & Budzier, 2011 · grid illustrative

We plan the best case



Reality takes the scenic route

Don’t ask how long
Ask how long it took

similar past projects · days



“Reference class forecasting”

Kahneman & Tversky · data illustrative

PARKINSON

Don't pad

HOFSTADTER

You'll be late anyway

Plan with your history, not your hopes

WHAT I TAKE FROM THIS

A big task is always a risk, so I plan it step by step:

1. **If we have built something similar, I start from how long it really took, not from what we estimated**
2. **For new work, we build a quick prototype before we estimate, when we can**
3. **I compare the team's estimate with mine to calibrate both**
4. **The plan has milestones. Each one says what must be ready by its date, so a delay shows early**
5. **After the first weeks of work, a stable core team shows its real pace, and we recalculate the forecast**

To stakeholders, I give two dates: the one we aim for and the one we promise

Sources: D. Hofstadter, "Gödel, Escher, Bach", 1979 · Buehler, Griffin and Ross, 1994 · Flyvbjerg and Budzier, 2011 · Kahneman and Tversky; Flyvbjerg

next: the 90-90 rule